

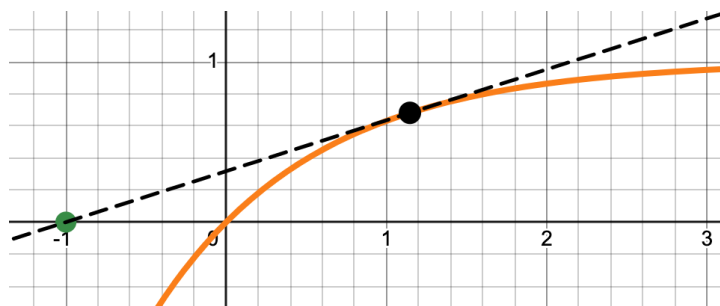
Problems

T₅ A bee goes from flower to flower to get pollen.

At some point it has to decide when it advantageous to move to a different flower.

Assume the following:

- Let t be the time in minutes since the bee landed in the current flower.
 - When the bee settles into a flower, it is full of pollen.
 - The bee starts gathering its pollen at a rate proportional to the amount of pollen left in the flower (with proportionality constant 1).
 - It takes 1 minute for the bee to find a new flower full of pollen.
- (a) Find a function for the amount of pollen the bee has gathered since it landed on a flower.
Hint. Keep track of the fraction of pollen still in the flower.
- (b) Calculate the average rate of pollen gathering by the bee starting from the time the bee left the previous flower until time t .
- (c) The bee decides to move on to the next flower when the average rate of pollen gathering starts decreasing. At what time does that happen?
- (d) Consider the following figure



Explain how it relates to this problem.

T₆ Let us find a formula for the air density $\rho(y)$ at an altitude of y metres.

(a) Consider the following two premises:

(P₁) Air density ρ (in kg/m³) is proportional to air pressure p (in Pa), with proportionality constant $\frac{M}{RT}$, where:

- M = molar mass of dry air = 0.0289644 kg/mol
- R = ideal gas constant = 8.31447 J/(mol·K)
- T = temperature of the air (assuming it is constant)

(P₂) Air pressure p is the weight of the air above

For each premise, find a relation the air density with the air pressure.

(b) Using these two relations, obtain an equation for the air density $\rho(y)$. Then obtain a differential equation for the air density $\rho(y)$.

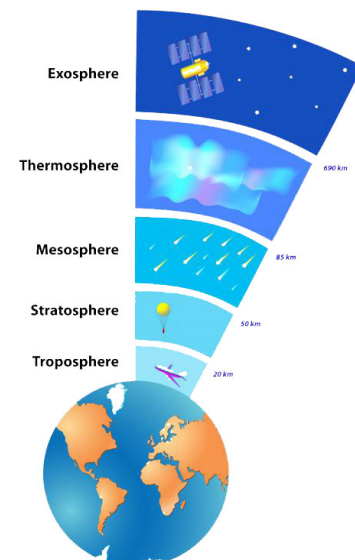
(c) Solve this differential equation to obtain a formula for the air density $\rho(y)$ assuming that $\rho(y_0) = \rho_0$.

(d) Use this formula to calculate the mass of Earth's atmosphere.
Here are some (simplified) facts that will help you:

- The **troposphere** and **stratosphere** contain 99.9% of the mass of the atmosphere
- The temperature in the **troposphere** decreases linearly with altitude and
 - T_0 = standard temperature at sea level = 288.15 K
 - L = temperature lapse = 0.0065 K/m
- Air pressure in the **troposphere** is given by

$$p_T(h) = p_0 \left(1 - \frac{Lh}{T_0} \right)^{\frac{gM}{RL}}$$

where h = altitude above sea level and p_0 = is the standard air pressure at sea level = 101.325 kPa



- The temperature in the **stratosphere** is constant.

(e) It is often said that the Troposphere contains three quarters of the atmosphere's mass. Using the International Standard Atmosphere,¹ check whether this model satisfies this.

(f) Compare this to the actual mass of the atmosphere. How can it be improved?

¹https://en.wikipedia.org/wiki/International_Standard_Atmosphere